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An open-source Python-based model to represent the charging flexibility and vehicle-to-grid potential of electric vehicles in energy systems models: venco.py

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Summary

The bottom-up simulation model venco.py provides boundary conditions for load shifting and vehicle-to-grid (V2G) potentials based on mobility demand data and techno-economic assumptions. The tool allows the modelling of the energy demand and flexibility potential of electric vehicle (EV) fleets within the field of energy systems analysis. It can be used to model EV charging both in a controlled (load shifting and V2G) and uncontrolled manner. Different charging infrastructure assumptions as well as different technical characteristics for the vehicle fleet can be considered, which allow the modelling of heterogeneous fleets and various infrastructure deployment scenarios. The main modelling outputs include battery drain profiles, charging capacity profiles, a minimum and a maximum battery energy levels and uncontrolled charging profiles both at single vehicle and at fleet level. The first four profiles can be used as constraining boundaries in other models to determine optimal charging strategies for the vehicles, representing this way an endogenous EV demand, whereas the last profile simulates a case, where charging is uncontrolled and the vehicles can charge as soon as a charging possibility is available. The very generic nature of the model's resulting profiles allows their usage to answer a wide range of research questions in multiple models, such as energy system optimisation models (Brown et al., 2018; Howells et al., 2011; Wetzel et al., 2024) or agent-based electricity market models (Schimeczek et al., 2023).

Statement of need

Being able to estimate the electricity demand, the load shifting or V2G potential of EV fleets is of interest in a wide range of research fields and industry applications, be it to analyse future grid ancillary service demands, the profit of aggregators managing EV fleets, to calculate the additional electricity demand caused by an increased penetration of EVs or the demand-side flexibility these could provide to the energy system.

Within the energy systems analysis field, two approaches to model EVs can be identified, being either data-driven approaches based on empirically measured data or bottom-up simulations. Data-driven approaches start from either measured state of charge data on board of the vehicles or data collected at charging stations to then scale these up to represent a fleet with respect to the modelling scope (Märtz et al., 2022). On the other hand, bottom up simulation models derive the flexibility and load profiles of EV fleets by taking a set of assumptions related to the charging controllability and the technical parameters of the vehicles (Gaete-Morales et al., 2021). These models usually use mobility pattern data as a starting basis, which can be taken from National Travel Surveys (NTSs) or be the simulation results from transport

research models. A few tools to calculate EV fleet profiles have been recently published, such as Emobpy (Gaete-Morales et al., 2021), RAMP-mobility (Mangipinto et al., 2022), OMOD (Strobel & Pruckner, 2023), SimBEV (*SIMBEV Documentation*, n.d.) and MobiFlex (Märtz et al., 2022). Similarly to these models, venco.py was developed to provide demand and flexibility potentials of future EV fleets. While the application focus of venco.py to date was on plug-in battery electric vehicles, the novelty of the framework in contrast to existing models consists in its upcoming capability to also model plug-in hybrid electric vehicles (PHEV) and in ready to use interfaces to several existing NTSs for European countries. Another novelty consists in the possibility to carry out analyses for specific vehicle classes, socio-economic groups or higher spatial resolutions than national level. As NTSs contain a broad set of data in addition to the mere mobility patterns, it is possible to generate EV profiles that can be linked to specific groups of EV users.

The venco.py model is completely developed in Python and openly available on Gitlab at https://gitlab.com/dlr-ve/esy/vencopy/vencopy.

Modelling approach

The venco.py model is designed to allow the modelling of heterogeneous vehicle fleets in a user-friendly and flexible way. While the model has so far been applied on the German NTS ("Mobilität in Deutschland") (infas, DLR, IVT und infas 360, 2018), the English NTS ("National Travel Survey: England 2019") (Department for Transport, 2019), the Dutch NTS ("Documentatie onderzoek Onderweg in Nederland 2019 (ODiN2019)") (Centraal Bureau voor de Statistiek, 2022) and the French NTS ("La mobilit'e locale et longue distance des Français - Enqu^ete nationale sur la mobilit'e des personnes en 2019") (Le service des données et études statistiques, 2023), it can accommodate any input data representing mobility patterns of a fleet for the respective modelling scope because of its flexible parsing approach. The model additionally features a user-defined temporal and geographical resolution, and allows the modelling both at an individual vehicle and at fleet level.

The model is based on the main building blocks shown in Fig. 1. Starting from a parsing interface to mobility datasets, the data is first cleaned and filtered for plausibility and variables are consolidated to model internal variables. Following that, the charging infrastructure allocation takes place, which makes use of a basic charging infrastructure model, that assumes the availability of charging stations when vehicles are parked. Since the analytical focus of the model lies on a regional level, the charging infrastructure availability is allocated either based on binary mappings related to the respective trip purpose, or it can be based on probability distributions for charging availability. This way, different charging availability scenarios can be distinguished. Subsequently, the demand and flexibility estimation are calculated, which are based on techno-economic input assumptions related mostly to the vehicle battery capacity and the power consumption. After an iteration process, this yields the minimum and maximum battery constraints. The individual trips at the survey day are afterwards consolidated into person-specific travel diaries, which comprise multiple trips that are carried out by each vehicle within a day. These daily activities are then discretised from a table to a time series format. The last step in the framework is the aggregation from single vehicle profiles to fleet level and the creation of annual profiles from the daily and weekly samples, with an option to normalise the profiles or scale them to a desired fleet size.

The model output is furnished with an automatic metadata generation in line with the metadata requirements of the Open Energy Platform (OEP) (Booshehri et al., 2021; Hülk et al., 2024), which enables the fulfilment of the FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson et al., 2016).

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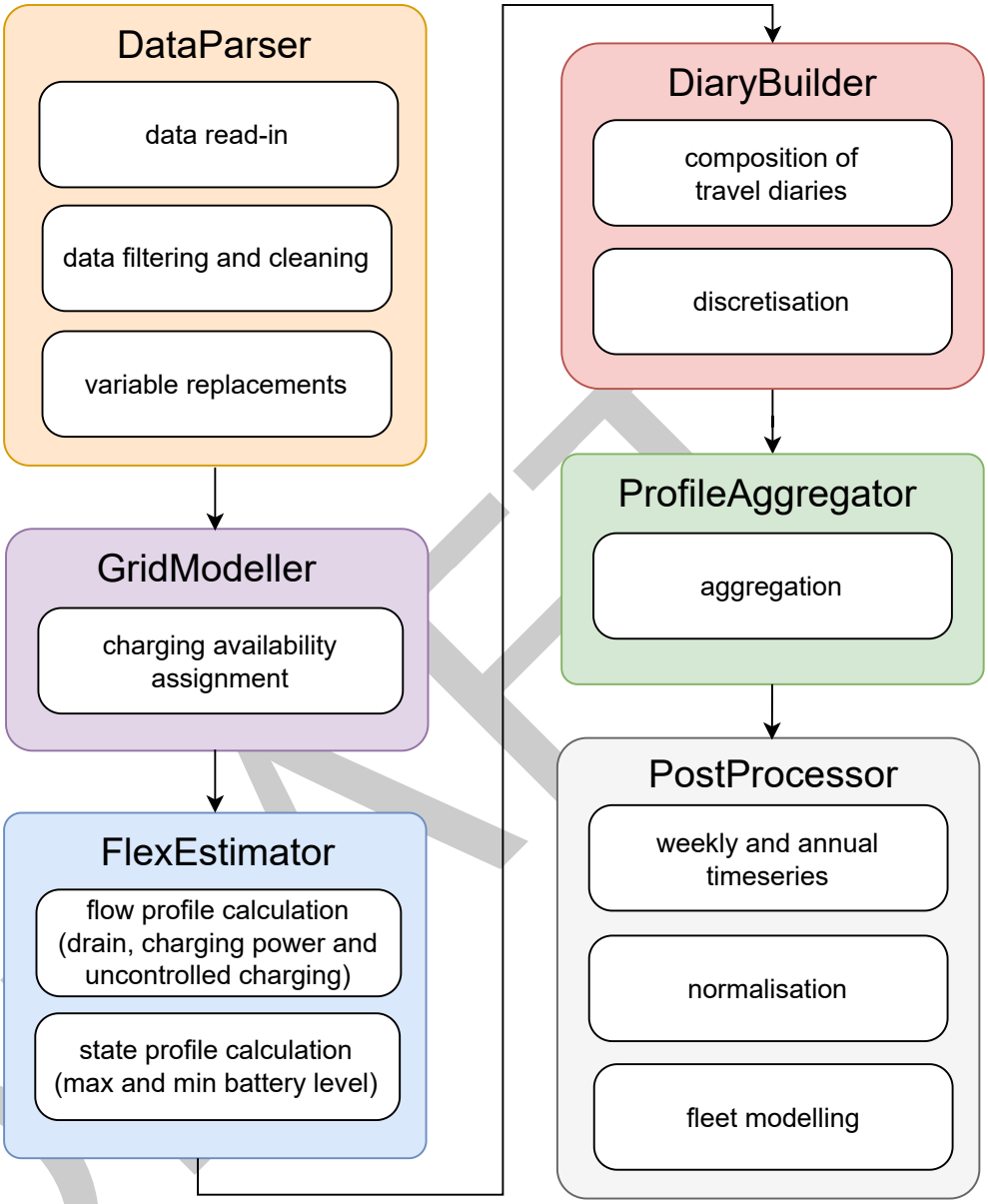

Figure 1: Structure of venco.py.

Projects and publications

Earlier versions of venco.py have been used throughout different projects. In the EVer project (*EVer Project Description*, n.d.), the model was, for example, used to perform a comparison of different modelling approaches for demand-side flexibility assets in the energy system optimisation model REMix (Wetzel et al., 2024). A more detailed assessment of the transport sector has been carried out in the BEniVer (*BENiVer Project Description*, n.d.) and UrMoDigital (*UrMoDigital Project Description*, n.d.) projects, where the role of synthetic fuels, EVs and new vehicle concepts have been analysed respectively. The model has also been used within the framework of the SEDOS project (*SEDOS Project Description*, n.d.), whose aim is the creation of a an open-source national reference energy system dataset and the development of a reference model for three open-source frameworks (oemof (Krien et al., 2020), TIMES

(*The Integrated MARKAL-EFOM System (TIMES) - a Bottom-up Optimization Model for Energy-Environment Systems*, n.d.), FINE (Groß et al., 2023)) with a focus on the German energy system. In the En4U project (*En4U Project Description*, n.d.) venco.py was used to create representative load profiles based mobility patterns of different household clusters and on residential time varying tariffs. In the European DriVe2X project the model is being used to assess the impact of mass deployment of V2X technologies on the energy systems as a whole (*DriVe2X Project Description*, n.d.).

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